

Partical 04

```
hanish@Hanish:~/2520090107_OSSP$ nano wait_process.c
hanish@Hanish:~/2520090107_OSSP$ |
```

```
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main()
{
    pid_t pid1, pid2;

    pid1 = fork();

    if (pid1 == 0)
    {
        printf("Child 1 PID: %d\n", getpid());
        sleep(3);
        exit(0);
    }

    pid2 = fork();

    if (pid2 == 0)
    {
        printf("Child 2 PID: %d\n", getpid());
        sleep(5);
        exit(0);
    }

    wait(NULL);

    printf("First child completed.\n");

    waitpid(pid2, NULL, 0);

    printf("Second child completed.\n");

    return 0;
}
```

```
hanish@Hanish:~/2520090107_OSSP$ gcc wait_process.c -o wait_process
hanish@Hanish:~/2520090107_OSSP$ ./wait_process
Child 2 PID: 1252
Child 1 PID: 1251
First child completed.
Second child completed.
hanish@Hanish:~/2520090107_OSSP$ |
```

Part 2 - Zombie process

```
hanish@Hanish:~/2520090107_OSSP$ nano zombie.c
hanish@Hanish:~/2520090107_OSSP$ |
```

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int main()
{
    pid_t pid = fork();

    if (pid == 0)
    {
        printf("Child process\n");
        exit(0);
    }
    else
    {
        printf("Parent process\n");
        sleep(30);
    }

    return 0;
}
```

```
hanish@Hanish:~/2520090107_OSSP$ nano zombie.c
hanish@Hanish:~/2520090107_OSSP$ gcc zombie.c -o zombie
hanish@Hanish:~/2520090107_OSSP$ ./zombie
Parent process
Child process
hanish@Hanish:~/2520090107_OSSP$ |
```

```
hanish@Hanish:~$ cd ~/2520090107_OSSP
hanish@Hanish:~/2520090107_OSSP$ ps -el | grep zombie
hanish@Hanish:~/2520090107_OSSP$ |
```